

35—BAR SCALES (continued)

BAR SCALE CALCULATIONS — KILOMETERS (1 kilometer = 100,000 centimeters)								
FRACTIONAL SCALE	SCALE TO MAP REPRESENTATION		TO FIND KILOMETERS PER CENTIMETER (CM) (x in ratio)	KILOMETERS PER CM	TOTAL KILOMETERS ON SCALE	TO FIND TOTAL SCALE LENGTH IN CENTIMETERS (y in ratio)	TOTAL SCALE LENGTH IN	
	Scale Unit :	represents Map Unit	Use ratio below or $\frac{\text{SCALE}}{100\,000}$			Use ratio below or Kilometers on scale Kilometers per cm	CENTI-METERS	MILLI-METERS
1:12 000	1 cm	12 000 cm	$\frac{100\,000}{1} = \frac{12\,000}{x}$	0.120	1.5	$\frac{0.120}{1} = \frac{1.5}{y}$	12.500	125.00
1:20 000	1 cm	20 000 cm	$\frac{100\,000}{1} = \frac{20\,000}{x}$	0.200	2	$\frac{0.200}{1} = \frac{2}{y}$	10.000	100.00
1:24 000	1 cm	24 000 cm	$\frac{100\,000}{1} = \frac{24\,000}{x}$	0.240	2	$\frac{0.240}{1} = \frac{2}{y}$	8.333	83.33
1:25 000	1 cm	25 000 cm	$\frac{100\,000}{1} = \frac{25\,000}{x}$	0.250	2	$\frac{0.250}{1} = \frac{2}{y}$	8.000	80.00
1:31 250	1 cm	31 250 cm	$\frac{100\,000}{1} = \frac{31\,250}{x}$	0.3125	2	$\frac{0.3125}{1} = \frac{2}{y}$	6.400	64.00
1:31 680	1 cm	31 680 cm	$\frac{100\,000}{1} = \frac{31\,680}{x}$	0.3168	2	$\frac{0.3168}{1} = \frac{2}{y}$	6.313	63.13
1:48 000	1 cm	48 000 cm	$\frac{100\,000}{1} = \frac{48\,000}{x}$	0.480	6	$\frac{0.480}{1} = \frac{6}{y}$	12.500	125.00
1:50 000	1 cm	50 000 cm	$\frac{100\,000}{1} = \frac{50\,000}{x}$	0.500	6	$\frac{0.500}{1} = \frac{6}{y}$	12.000	120.00
1:62 500	1 cm	62 500 cm	$\frac{100\,000}{1} = \frac{62\,500}{x}$	0.625	8	$\frac{0.625}{1} = \frac{8}{y}$	12.800	128.00
1:63 360	1 cm	63 360 cm	$\frac{100\,000}{1} = \frac{63\,360}{x}$	0.6336	8	$\frac{0.6336}{1} = \frac{8}{y}$	12.626	126.26
1:72 000	1 cm	72 000 cm	$\frac{100\,000}{1} = \frac{72\,000}{x}$	0.720	8	$\frac{0.720}{1} = \frac{8}{y}$	11.111	111.11
1:75 000	1 cm	75 000 cm	$\frac{100\,000}{1} = \frac{75\,000}{x}$	0.750	8	$\frac{0.750}{1} = \frac{8}{y}$	10.666	106.66
1:96 000	1 cm	96 000 cm	$\frac{100\,000}{1} = \frac{96\,000}{x}$	0.960	9	$\frac{0.960}{1} = \frac{9}{y}$	9.375	93.75
1:100 000	1 cm	100 000 cm	$\frac{100\,000}{1} = \frac{100\,000}{x}$	1.000	11	$\frac{1.000}{1} = \frac{11}{y}$	11.000	110.00
1:125 000	1 cm	125 000 cm	$\frac{100\,000}{1} = \frac{125\,000}{x}$	1.250	12	$\frac{1.250}{1} = \frac{12}{y}$	9.600	96.00
1:150 000	1 cm	150 000 cm	$\frac{100\,000}{1} = \frac{150\,000}{x}$	1.500	12	$\frac{1.500}{1} = \frac{12}{y}$	8.000	80.00

To find kilometers per centimeter on 1: 12 000 map ...**Solution ...**

$$100\,000 \cdot x = 12\,000 \cdot 1$$

100 000 centimeters = 1 kilometer
Show in ratio as ...

$$\frac{100\,000}{1} \frac{\text{centimeters}}{\text{kilometers}}$$

Let SCALE (12 000) be in centimeters
Fractional scale says 1 centimeter represents
12,000 centimeters
Let x be kilometers that 1 cm represents on map
Show in ratio as ...

$$\frac{12\,000}{x} \frac{\text{centimeters}}{\text{kilometers}}$$

$$\frac{100\,000}{1} = \frac{12\,000}{x}$$

$$\frac{100\,000}{100\,000} x = \frac{12\,000}{100\,000}$$

$$x = \frac{12\,000}{100\,000} \text{ (SCALE)}$$

$$x = 0.120$$